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Diffusion of molybdenum in α -iron

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Abstract

Tracer diffusion coefficient of ⁹⁹Mo in α -iron has been determined in the temperature range between 833 and 1163 K by use of a serial sputter-microsectioning technique. The temperature dependence of the diffusion coefficient, D , in the whole temperature range of α -iron across the Curie temperature can be expressed by

$$D = \left(1.48^{+1.51}_{-0.75} \right) \times 10^{-2} \exp \left[-\frac{(282.6 \pm 6.4 \text{ kJ mol}^{-1})[1 + (0.074 \pm 0.005)s^2]}{RT} \right] \text{ m}^2 \text{ s}^{-1}$$

where s is the ratio of spontaneous magnetization at T K to that at 0 K. The factor 0.074 in the above equation is smaller than 0.156 for self-diffusion in α -iron, showing that the influence of the magnetic transformation on the diffusion of molybdenum in α -iron is smaller than that on the self-diffusion as expected from the negative magnetic moment of molybdenum in the lattice of the ferromagnetic iron. © 2002 Acta Materialia Inc. Published by Elsevier Science Ltd. All rights reserved.

Keywords: Diffusion; Magnetic properties; Molybdenum; Iron

1. Introduction

Molybdenum is an important alloying element to iron-base alloys for improvement in their strength and heat resistance capacity. It is well-known that a small addition of molybdenum remarkably increases creep strength of the steels [1]. Atomic movement plays a key role in creep deformation at high temperatures. Therefore, the

experimental data on the self-diffusion of both components in dilute iron-molybdenum alloys are essential to evaluate their usefulness as a structural material at high temperatures. However, the appropriate diffusion data on iron-molybdenum alloys are quite sparse [2–4].

In the past, tracer impurity diffusion coefficient of molybdenum in α -iron has been measured by Borisov et al. [5], Kučera et al. [6] and Nohara and Hirano [7] using the residual activity method. Alberry and Haworth [8] used the diffusion couple method to determine the diffusion coefficient of molybdenum in α -iron. However, the numerical values of the diffusion coefficients reported by

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these authors exhibit wide variation. Nohara and Hirano [7] have measured the diffusion coefficient in the paramagnetic α -iron and also in the ferromagnetic α -iron up to 50 K below the Curie temperature (1043 K), while Borisov et al. [5], Kučera et al. [6] and Alberry and Haworth [8] have concentrated in the measurements in the temperature range above the Curie temperature. On the other hand, in the application of ferritic iron alloys for thermal power plants, the reliable experimental data at lower temperatures in particular around 900 K is necessary.

The precise determination of the diffusion coefficient of an atom in solids is commonly made by the use of radio-tracer method in combination with an appropriate serial sectioning technique to establish the penetration profiles of the diffusing tracer in the specimen. It is also essential to use, as far as possible, high purity single crystal specimen with stress free surface [9]. Alternatively, specimen with large grain size should be employed. Otherwise, apparent diffusion coefficient affected by impurities and short-circuiting diffusion paths such as dislocations and grain boundaries are recorded. The influence of short circuit paths on diffusion is more serious at lower temperatures. In recent studies on self-diffusion in α -iron [4,10,11], it has been noted that the self-diffusion coefficient measured by the sputter-microsectioning [10,11] is smaller than that obtained by the use of microtome-sectioning [11,12]. In the latter experiments, the mean penetration distance, $2\sqrt{Dt}$ (D =diffusion coefficient, t =diffusion time), is in the range 30–100 μm which is significantly greater than inter-dislocation spacing in the specimen. As a result, the measured self-diffusion coefficients in α -iron obtained by the conventional mechanical sectioning techniques are usually affected by the enhancement effect due to the diffusion along the dislocation networks [4,10,11]. So far, diffusion of chromium [13], cobalt [14,15] and nickel [16] in α -iron has been studied by use of high purity well-annealed specimens and the radio-tracer method combined with microsectioning techniques. Recently, the Rutherford backscattering spectrometry and heavy ion Rutherford backscattering spectrometry have also been used to analyze the

submicron penetration profiles arising from the diffusion of tin in α -iron [17].

Below the Curie temperature, the Arrhenius plot of self-diffusion coefficients in the ferromagnetic state of iron deviates downwards from the Arrhenius relationship extrapolated from the paramagnetic state [4,10,11]. The equations describing such temperature dependence of the diffusion coefficients in the whole temperature range across the Curie temperature have been proposed by several authors such as Ruch et al. [18], Hettich et al. [19], Braun and Feller-Kniepmeier [20], Čermák et al. [16] and Mirani et al. [21]. The equation given by Ruch et al. [18] is

$$D = D_0^p \exp[-Q^p(1 + \alpha s^2)/RT] \quad (1)$$

Here D_0^p and Q^p are, respectively, the preexponential factor and the activation energy for the self-diffusion in the paramagnetic state. The value of s , the ratio of the spontaneous magnetization at T K to that at 0 K, has been experimentally determined by Potter [22] and Crangle and Goodman [23]. The constant α , expressing the extent of the influence of the magnetic transformation on diffusion, is given by

$$\alpha = \frac{\alpha_f + \alpha_m}{Q^p}, \quad (2)$$

where α_f and α_m are the increments of formation energy and migration energy, respectively, of a vacancy by the magnetic transformation. α_f is related to the exchange integral J between the dithering atom and its nearest neighbors in the equilibrium position. Let z be the coordination number, then $\alpha_f = zJ/2$, and estimated to be 4.3 kJ mol^{-1} for iron by Ruch et al. [18]. α_m is related to the exchange integral J^* between a jumping atom in the saddle point and a nearest neighbor atom as $\alpha_m = cJ^*$ where c is a constant. Theoretical calculation of α_m is not feasible at present because of difficulty in the estimation of J^* . In the model by Ruch et al. [18], the short range magnetic spin ordering has not been taken into consideration. On the other hand, its effect has been emphasized in the models proposed by Hettich et al. [19], Braun and Feller-Kniepmeier [20], Čermák et al. [16] and Mirani et al. [21]. However, the self-diffusion coefficient [10] and the solute

diffusion coefficient of chromium [13] in α -iron obtained by the present research group have shown that the Arrhenius plot, from the Curie temperature to the α - γ transformation temperature is linear without any influence of the short range magnetic spin ordering, whereas, the Arrhenius plot for the diffusion of cobalt in α -iron [15] in the same temperature region approaches the linear Arrhenius line for the self-diffusion with increase of temperature in an asymptotic manner. This means that a local magnetic field around a solute cobalt atom is stronger than the magnetic field of iron matrix and remains so even above the Curie temperature. On the other hand, around a solute chromium atom, it is weaker than the magnetic field of iron matrix and nearly diminishes at the Curie temperature. The larger magnetic field around a diffusing cobalt atom is related to the positive change in magnetization ($+1.0 \mu_B/\text{atom}$) of α -iron on alloying with cobalt [24,25], whilst, the smaller magnetic field around a diffusing chromium atom is related to the negative change ($-2.6 \mu_B/\text{atom}$) due to the presence of chromium [25]. Furthermore, negative change in magnetization ($-2.3 \mu_B/\text{atom}$) [25] of α -iron is also expected on alloying with molybdenum. Thus, because of the way molybdenum influence the magnetic moment in α -iron, the diffusion behavior of molybdenum in α -iron above and below the Curie temperature presents an interesting comparison with those of cobalt and chromium and considered worthy of fresh investigation.

The purpose of this work is to study the diffusion behavior of molybdenum in high purity α -iron in a wide range of temperature by use of the sputter-microsectioning and radioactive tracer ^{99}Mo to elucidate the extent of magnetic moment in the diffusion behavior of α -iron. This work is a part of experimental studies on self-diffusion of both components in dilute iron-molybdenum alloys.

2. Experimental procedure

High purity electrolytic iron, “Miron HP” supplied by Toho Zinc Co., was induction-melted in a cold-copper crucible under 2×10^{-4} Pa after

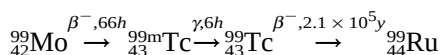
evacuation of the chamber to a base pressure of 10^{-7} Pa. The ingot was hot-forged and machined in the form of a rod 12 mm in diameter. The chemical analysis of the material is shown in Table 1. The rod was cut to make the disc specimen 1 mm in thickness. The flat face of the specimen was ground on abrasive papers and electropolished at 273 K in an aqueous solution containing 77 vol.% acetic acid and 18 vol.% perchloric acid. To induce the grain growth and reduce the presence of carbon, nitrogen and oxygen, the specimen was annealed in the δ -iron phase at 1723 K for 36 h in a stream of hydrogen gas purified by permeation through a palladium tube at 700 K [10]. The resultant grain size was about 3–5 μm . After the grain growth, the specimen was ground on abrasive papers and polished with fine alumina paste and electropolished under the same conditions as mentioned above. To obtain a stress-free surface, the specimen was further annealed at 1123 K for 5.4 ks in a stream of the purified hydrogen gas.

The radioisotope ^{99}Mo (β -rays, 0.448 MeV–17% and 1.234 MeV–82%; half-life, 65.84 h) was supplied in the form of Na_2MoO_4 in 0.11 kmol m^{-3} NaOH solution by Japan Atomic Energy Research Institute. Taking care to avoid oxidation of the sample, the radioisotope was electroplated on the surface of the specimen with 14 mA cm^{-2} for 0.5–2.0 min. The specimens were then sealed in quartz tubes under 10^{-4} Pa and diffused at a temperature range from 833 to 1163 K for 1.5 to 90 ks in furnaces preset at the desired temperatures and controlled to within ± 0.5 K. To measure the penetration profiles of the diffusing species into the specimen, two types of the serial sputter-microsectioning apparatus were used; one is the radio-frequency type for long penetration profiles in specimen diffused at high temperatures and the other is the ion-beam type for short penetration profiles at lower temperatures. The sputtering rate of the former is 50–80 nm min^{-1} , whereas that of latter is 5–8 nm min^{-1} . Thus, radio-frequency technique is

Table 1
Chemical analysis of iron (mass ppm)

C	N	O	P	S	Cr	Ni	Si
9	<5	14	<1	<1	28	7	0.2

appropriated at higher temperatures whereas ion-beam method is better suited for application at lower temperatures due to smaller penetration depth. The details of the methods are described elsewhere [26,27]. For each specimen, 15–35 successive sections were sputtered. A constant fraction more than 60% of the sputtered-off material was collected on an aluminum foil. The intensity of the radioactivity of the isotope in each section was measured by a well-type Tl-activated NaI detector with a 1024 channels pulse height analyzer. The radioactive diffusant ^{99}Mo is a β^- -emitter which decays through the reaction



via the radioactive daughter $^{99\text{m}}\text{Tc}$ (γ -rays, 0.142 MeV–80%; half-life, 6.01 h). For reasons of efficiency the 0.142 MeV γ -peak of $^{99\text{m}}\text{Tc}$ was counted. After a sufficiently long time gap after the diffusion anneal (at least ten half-lives of $^{99\text{m}}\text{Tc}$), $^{99\text{m}}\text{Tc}$ activity is proportional to the concentration of ^{99}Mo [28]. ^{99}Ru decay product on the final count is negligible because the NaI detector used for counting is programmed to record the γ -rays coming from $^{99\text{m}}\text{Tc}$ in a selective manner. The β -rays coming from all other isotopes are easily absorbed and do not influence the counts.

3. Results

For one-dimensional volume diffusion of a tracer from an infinitesimally thin surface layer into a sufficiently long rod, the solution of Fick's second law is given by

$$I(X,t) \propto C(X,t) = [M/\sqrt{\pi Dt}] \exp(-X^2/4Dt) \quad (3)$$

where $I(X,t)$ and $C(X,t)$ are respectively the intensity of the radioactivity and the concentration of the tracer at a distance X from the original surface after a diffusion time t . D is the volume diffusion coefficient of the tracer and M is the total amount of tracer deposited on the surface before the diffusion.

Figs. 1 and 2 show the plots of $\ln I(X,t)$ vs. X^2 for the diffusion of ^{99}Mo in α -iron at higher and

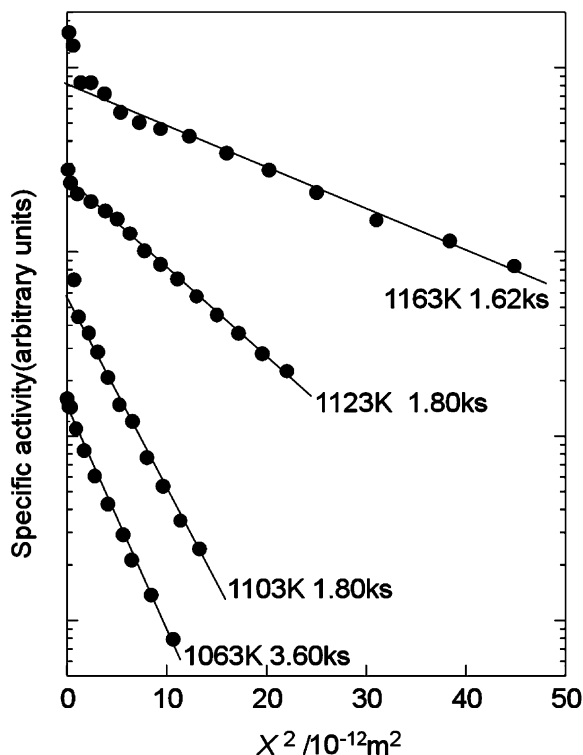


Fig. 1. Example penetration profiles for diffusion of ^{99}Mo in paramagnetic state of α -iron.

lower temperatures, respectively. In accordance with Eq. (3), all the plots in Figs. 1 and 2 are linear, indicating that the concentration profiles are governed by the volume diffusion. The diffusion coefficients calculated from the slope of the plot are listed in Table 2, and their Arrhenius plots are shown in Fig. 3 along with the data of Borisov et al. [5], Kučera et al. [6], Alberry and Haworth [8] and Nohara and Hirano [7]. The temperature dependence of the self-diffusion coefficient in α -iron [10,11,19] is also plotted in Fig. 3. The self-diffusion coefficients obtained by Iijima et al. [10], Hettich et al. [19] and Lübbehusen and Mehrer [11] are good in agreement with each other and hence, other experimental data points are not shown in Fig. 3 to avoid the cluttering of data points. It is seen that the diffusion coefficients of molybdenum determined by the present work and by other authors [5,6,8] are larger than the self-diffusion coefficients. However the results obtained by Nohara and Hirano [7] are smaller than

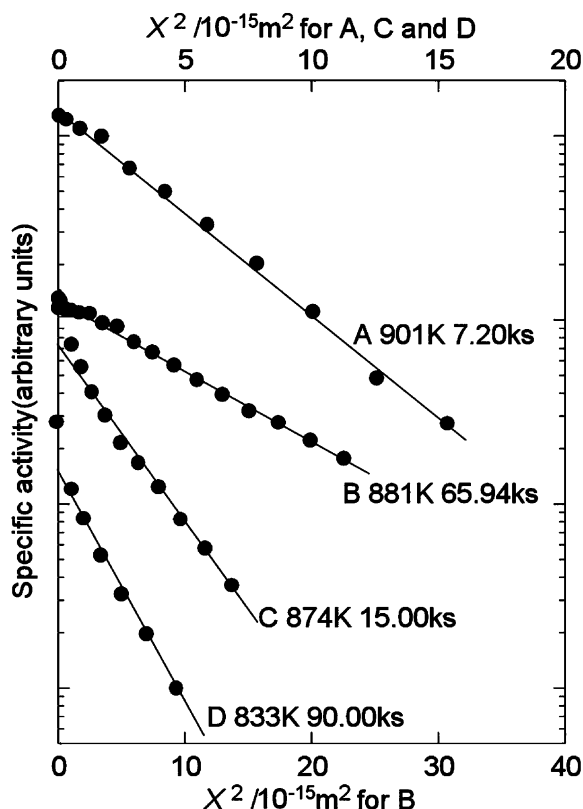


Fig. 2. Example penetration profiles for diffusion of ^{99}Mo in ferromagnetic state of α -iron.

those of other authors and lie close to the self-diffusion coefficients. The diffusion coefficients of molybdenum in α -iron by the present work show a linear Arrhenius behavior between the Curie temperature ($T_c=1043\text{K}$) and the α - γ transformation temperature ($T_{\alpha-\gamma}=1184\text{K}$) with the pre-exponential factor, $D_0^p, (1.48^{+1.51}_{-0.75}) \times 10^{-2} \text{ m}^2 \text{ s}^{-1}$, and the activation energy, Q^p , $(282.6 \pm 6.4) \text{ kJ mol}^{-1}$ without any sign of the influence of the short range magnetic spin ordering on the diffusion. In the ferromagnetic state the diffusion coefficients are smaller than those extrapolated from the paramagnetic state, as is the case for the self-diffusion.

4. Discussion

As shown in Fig. 3, the diffusion coefficients for molybdenum in α -iron obtained by Borisov et al.

[5], Kučera et al. [6] and Alberly and Haworth [8] are larger than those recorded in the present work. These higher values reported by earlier worker are probably caused by the enhancement through diffusion along dislocations giving extended penetration distances suitable for the conventional mechanical sectioning procedures. The b.c.c. phase of δ -iron melts at 1811 K and the diffusion in α -iron at $0.65T_m$ could be treated as low temperature diffusion in high temperature δ -iron. Usually, temperature induced phase transformation leads to the formation of more open structure at higher temperatures, e.g. h.c.p. or f.c.c. transforming to b.c.c. In the case of iron, it is just the reverse. Hence for the purpose of the present discussion, we can regard diffusion in the paramagnetic phase of α -iron as low temperature diffusion in b.c.c. δ -phase. Diffusion in the neighborhood of 0.5 – $0.6T_m$ is strongly influenced through short-circuiting along dislocation networks. We believe that this is the most likely cause of enhanced diffusion coefficient values reported by earlier authors [5,6]. Extended penetration profiles necessary to obtain the diffusion coefficients by interdiffusion techniques with the help of EPMA may also be probable cause of higher diffusion coefficients reported by Alberly and Haworth [8]. On the other hand, the D values obtained by Nohara and Hirano [7] are smaller than the values obtained in the present work as well all other previous workers. These authors employed residual activity method and counted γ -rays which are emitted at a low rate due to the β -decays of ^{99}Mo . This method is prone to yield inaccurate data for two reasons. In addition to the inherent problems associated with the residual sectioning techniques, low rate of γ emission gives rise to poor statistics in the radioactive assay of the concentration.

Table 3 summarizes the Arrhenius parameters for the self-diffusion [10] and the solute diffusion of cobalt [15], chromium [13] and molybdenum obtained by our group using the radioactive tracer method with serial sputter-microsectioning techniques. Fig. 4 shows a plot of the activation energy Q^p for self-diffusion [10], impurity diffusion of transition metals of Ni [16], Co [15], Cr [13], V [29], Ti [30] and Mo [present work] in the paramagnetic α -iron against $(r_s - r_{Fe})/r_{Fe}$, where r_s and

Table 2
Diffusion coefficient of molybdenum in α -iron

Temperature (K)	Diffusion time (ks)	Diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)	Technique
1163	1.62	2.75×10^{-15}	RF
1163	1.62	2.97×10^{-15}	RF
1143	1.50	1.64×10^{-15}	RF
1123	1.80	1.19×10^{-15}	RF
1103	1.80	6.57×10^{-16}	RF
1103	1.50	6.74×10^{-16}	RF
1088	3.00	3.95×10^{-16}	RF
1073	7.20	2.87×10^{-16}	RF
1063	3.60	1.84×10^{-16}	RF
1050	64.80	1.26×10^{-16}	RF
1041	3.60	8.81×10^{-17}	IBS
1033	3.60	5.37×10^{-17}	IBS
1023	68.40	3.74×10^{-17}	RF
993	3.60	9.07×10^{-18}	IBS
973	61.62	4.78×10^{-18}	RF
951	66.00	1.23×10^{-18}	RF
923	57.60	3.46×10^{-19}	RF
901	7.20	1.37×10^{-19}	IBS
881	65.94	4.87×10^{-20}	IBS
874	15.00	3.87×10^{-20}	IBS
853	64.80	1.19×10^{-20}	IBS
833	90.00	4.75×10^{-21}	IBS

RF, radio frequency sputter-microsectioning.

IBS, ion beam sputter-microsectioning

r_{Fe} are the radii of the solute atom for the coordination number 8 and of the iron matrix lattice [31]. A linear relationship is observed. Therefore, atomic size effect seems predominant in affecting the activation energy for the diffusion of transition metal solutes in the paramagnetic α -iron. The twin facts that the molybdenum is an effective alloying element for solid solution hardening of b.c.c iron and the large difference in the activation energies between self-diffusion and diffusion of molybdenum in the paramagnetic α -iron can both be ascribed to the size effect.

Eq. (1) can be rewritten as follows

$$\ln \frac{D(T)}{D_0^p} = -\frac{Q^p}{R} - \left(\frac{\alpha Q^p}{R} \right) s^2. \quad (4)$$

Substituting the values of $D(T)$ and D_0^p from the present work and the empirical value of $s(T)$ for pure iron by Potter [22] and Crangle and Goodman [23] into Eq. (4), $\ln[D(T)/D_0^p]$ was calculated as a function of s^2 and plotted in Fig. 5. A linear

relationship between $\ln[D(T)/D_0^p]$ and s^2 in Fig. 5 shows that in the ferromagnetic state of iron, the temperature dependence of diffusion coefficient of molybdenum obeys Eq. (1). From the slope and intercept of the straight line in Fig. 5, the values of α and Q^p were determined and listed in Table 3. These are compared with the values for self-diffusion [10], diffusion of chromium [13] and cobalt [15] in α -iron. Q^p calculated in this way is denoted by Q_{cal}^p . It is noted that the value of Q^p determined directly from the Arrhenius plot of $D(T)$ in the paramagnetic state is in good agreement with Q_{cal}^p , as shown in Table 3. Using the values of α ($=0.074$), D_0^p ($=1.48 \times 10^{-2} \text{ m}^2 \text{ s}^{-1}$) and Q^p ($=282.6 \text{ kJ mol}^{-1}$) obtained by the present work, $D(T)$ is calculated, and its Arrhenius plot is drawn as a solid line in Fig. 3. As shown in Table 3, the value of α ($=0.074$) for the diffusion of molybdenum in α -iron is about 53% smaller than 0.156 for the self-diffusion [10], indicating that the effect of magnetic transformation in iron on the

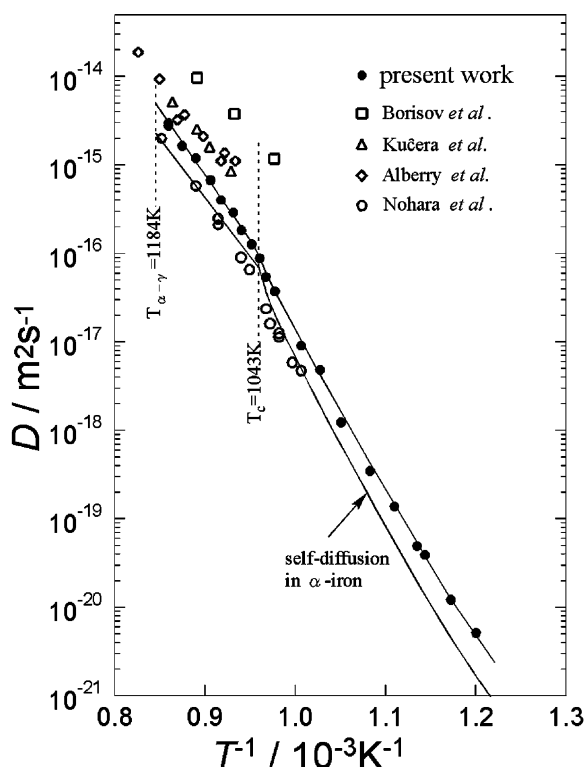


Fig. 3. Arrhenius plot of diffusion coefficients of molybdenum in α -iron in comparison with those by other authors. Arrhenius plot of the self-diffusion coefficient in α -iron is also shown.

Table 3

Preexponential factor, activation energy and constant α for self-diffusion and diffusion of molybdenum, chromium and cobalt in α -iron

	D_0^p ($\text{m}^2 \text{s}^{-1}$)	Q^p (kJ mol^{-1})	$Q_{\text{cal.}}^p$ (kJ mol^{-1})	α	$Q^f = Q^p(1 + \alpha)$ (kJ mol^{-1})
Self-diffusion	$2.76_{-1.04}^{+1.42} \times 10^{-4}$	250.6 ± 3.8	250.5 ± 0.4	0.156 ± 0.003	289.7 ± 5.1
Diffusion of molybdenum	$1.48_{-0.75}^{+1.51} \times 10^{-2}$	282.6 ± 6.4	283.1 ± 0.6	0.074 ± 0.005	303.5 ± 7.5
Diffusion of cobalt	2.76×10^{-4}	251	—	0.23	309
Diffusion of chromium	$3.73_{-1.40}^{+2.24} \times 10^{-3}$	267.4 ± 4.4	267.5 ± 0.5	0.133 ± 0.005	303.0 ± 6.3

solute diffusion of molybdenum is smaller than that on the self-diffusion. Using the value of Q^p ($=282.6 \text{ kJ mol}^{-1}$) for the diffusion of molybdenum in the paramagnetic iron, the activation energy Q^f ($= (1 + \alpha) Q_p$) in the ferromagnetic iron can be estimated to be $303.5 \text{ kJ mol}^{-1}$. Thus, the increment of the activation energy αQ^p by the magnetic transformation is 20.9 kJ mol^{-1} for the

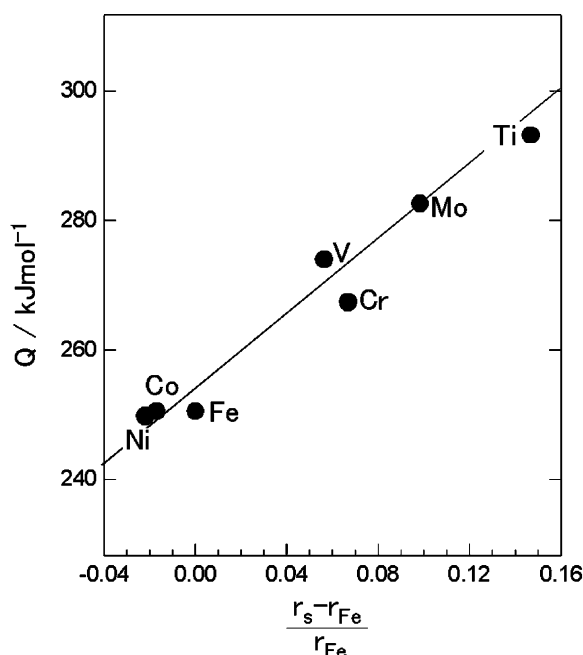


Fig. 4. Plot of Q^p vs. $(r_s - r_{\text{Fe}})/r_{\text{Fe}}$.

diffusion of molybdenum, while it is 39.1 kJ mol^{-1} for the self-diffusion. Thus in the paramagnetic iron, the activation energy for the diffusion of molybdenum is 32 kJ mol^{-1} larger than that for self-diffusion, whereas in the ferromagnetic region, this difference is reduced to only 13.8 kJ mol^{-1} . The

value of αs^2 can be taken as a measure of the degree of the influence of the magnetic transformation on diffusion. Putting $D(T)$, D_0^p and Q^p into Eq. (1), the value of αs^2 for diffusion of molybdenum is calculated as a function of temperature and shown in Fig. 6. The values of αs^2 for self-diffusion [10] and other solutes [13,15] are also shown in Fig. 6. In the case of self-diffusion and

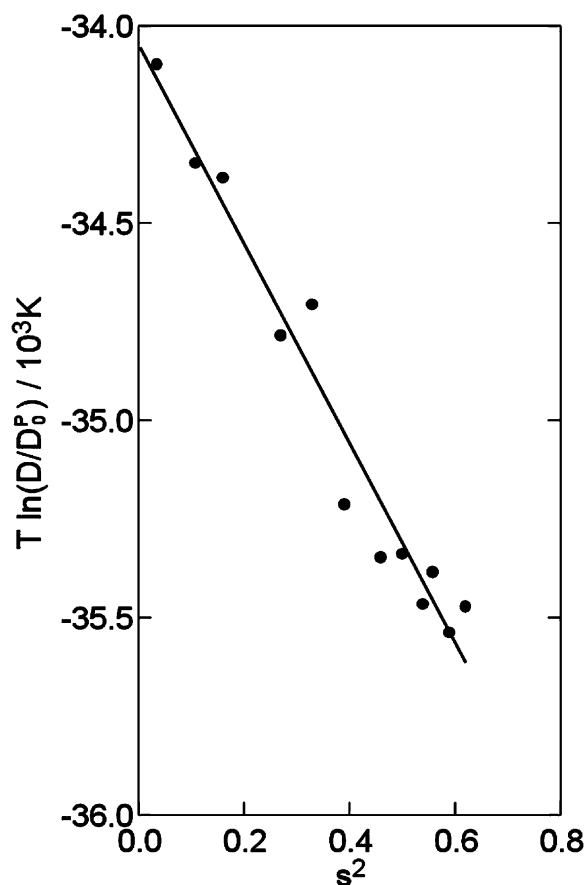


Fig. 5. Plot of $T \ln[D(T)/D_0^f]$ vs. s^2 for diffusion of molybdenum in α -iron of the ferromagnetic state.

diffusion of molybdenum and chromium in α -iron, αs^2 decreases gradually with increase of temperature and becomes zero at and above the Curie temperature of pure iron, although for the diffusion of molybdenum is about half of that for the self-diffusion. However, for the diffusion of cobalt in α -iron, αs^2 is larger than that for the self-diffusion in the whole temperature range and becomes zero at and above 1140 K which is 97 K higher than the Curie temperature of pure iron. This means that a local magnetic field around a solute cobalt atom is stronger than that of pure iron matrix and it remains so even above the Curie temperature. On the other hand, for molybdenum and chromium solute atoms, magnetic moment is weaker than that of the pure iron matrix and becomes zero at the Curie temperature. The experimentally determined

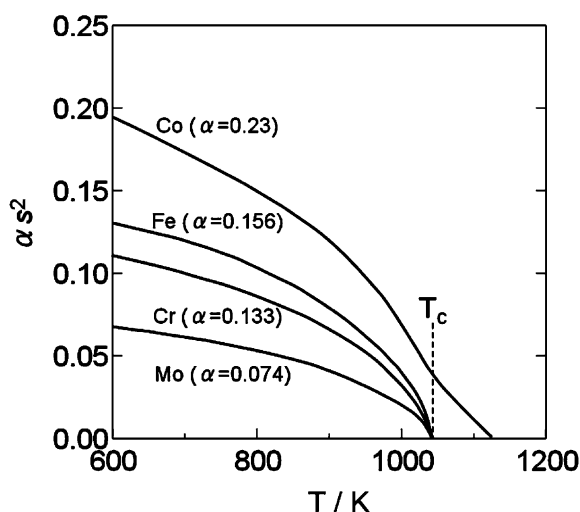


Fig. 6. Temperature dependence of αs^2 for diffusion of molybdenum in comparison with those for self-diffusion and diffusion of chromium and cobalt in α -iron.

values of the change in magnetization ΔM of α -iron on diluting with 3d, 4d and 5d elements are compiled by Pepperhoff and Acet [25]. As mentioned earlier, they are $+1.0 \mu_B/\text{atom}$ for cobalt, $-2.6 \mu_B/\text{atom}$ for chromium and $-2.3 \mu_B/\text{atom}$ for molybdenum. The value ΔM is made up of contribution arising from the influence of the impurity atoms on the magnetic moment of the surrounding iron atoms and from the contribution of the local impurity moment. The latter is the difference between the iron atom moment and the local impurity moment. Thus, ΔM is the sum of contributions from all neighbors extending out to several shells. According to theoretical calculations of ΔM for transition-metal impurities in iron by Drittler et al. [24], the values of ΔM are $+0.79 \mu_B$ for cobalt, $-3.29 \mu_B$ for chromium and $-3.31 \mu_B$ for molybdenum, although the local moments of impurities in iron are $+1.65 \mu_B$ for cobalt, $-1.29 \mu_B$ for chromium and $-0.52 \mu_B$ for molybdenum. Drittler et al. [24] have calculated the change of the total moment summed over the impurity and five shells of iron atoms. At present, it is difficult to connect the value of ΔM to the values of α_f and α_m in Eq. (2). However, it presents that the value of ΔM are related qualitatively to the order of magnitudes of αs^2 for the three solutes in Fig. 6.

5. Conclusion

Using the serial sputter-microsectioning technique with radioactive tracer ^{99}Mo , the diffusion coefficient of molybdenum in α -iron has been determined in the temperature range from 833 to 1163 K. The diffusion coefficient of molybdenum is two to three times larger than the self-diffusion coefficient in α -iron. The temperature dependence of the diffusion coefficient of molybdenum is influenced by the magnetic transformation in a way similar to that for the self-diffusion. The increase in activation energy due to the magnetic transformation for solute diffusion of molybdenum is found to be 20.9 kJ mol^{-1} which is smaller than the corresponding value of 39.1 kJ mol^{-1} for the self-diffusion of iron.

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